

TECHNICAL MEMORANDUM

TO: Elk Basin Irrigation District Board of Directors
FROM: S. M. Takahashi, P.E., D.GE Mountain West Dam Engineering
DATE: November 20, 2025
RE: Remediation Alternatives Evaluation — Elk Basin Dam FERC Project No. 4182
PROJECT: MWDE No. 2025-DAM-0034

1.0 PURPOSE

This memorandum evaluates remediation alternatives for the three conditions identified during the 2025 Periodic Facility Review (PFR) geotechnical investigation of Elk Basin Dam: (1) increasing seepage at the left abutment due to deterioration of the grout curtain in weathered granodiorite, (2) alkali-silica reaction (ASR) degradation of the arch dam concrete, and (3) ongoing creep displacement of the right abutment embankment. The evaluation presents feasible alternatives for each condition, with estimated costs, schedules, and recommended actions.

2.0 LEFT ABUTMENT — SUPPLEMENTAL GROUTING

2.1 Condition Summary

Left abutment seepage has increased from 12 gpm (2012) to 33.7 gpm (2025). The 2025 investigation confirmed the seepage path is through the weathered granodiorite zone (72 to 85 feet depth below the embankment crest), where the original 1967 grout curtain has lost effectiveness. Piezometer P-3 shows a correlated 71 percent pressure increase. The embankment clay core is intact with no piping. All seepage water is clear.

2.2 Alternatives Evaluated

Alternative A: Supplemental Curtain Grouting (Recommended)

A single row of grout holes drilled from the left abutment embankment crest, targeting the weathered granodiorite zone from 70 to 100 feet depth. Primary holes at 10-foot spacing with split-spacing (5-foot) as indicated by grout take. Balanced-stable grout mix (cement-bentonite) with real-time monitoring. Estimated 15 to 20 holes, 1,500 to 2,000 linear feet of drilling.

- **Cost:** \$180,000 to \$250,000
- **Schedule:** 4 to 6 weeks field work (summer 2026)
- **Effectiveness:** Expected to reduce seepage by 60-80%. Grout curtain renovation is well-established practice for weathered granitic foundations.
- **Risk:** Low. Hydrofracture risk is manageable with controlled injection pressures (rule of thumb: 0.5 psi/ft depth). No reservoir restriction needed.

Alternative B: Drainage Gallery Extension

Construct a drainage adit or horizontal drain array into the left abutment to capture and control seepage downstream of the core. This would not reduce seepage volume but would control pore pressures and provide permanent drainage.

- **Cost:** \$450,000 to \$650,000
- **Schedule:** 3 to 4 months
- **Effectiveness:** High for pore pressure control, but does not address root cause (grout curtain deterioration).
- **Risk:** Moderate. Excavation near embankment requires careful sequencing. Higher cost with no seepage reduction.

Alternative C: Monitoring Only

Enhance monitoring (weekly seepage, monthly piezometers) without physical intervention. Rely on turbidity sensors for early warning of internal erosion.

- **Cost:** \$15,000 per year (instruments and labor)
- **Effectiveness:** Does not address the condition, only detects changes. If seepage continues to increase at the current rate, it will reach approximately 50 gpm by 2035, which may trigger FERC enforcement action.

Recommendation: Alternative A (supplemental curtain grouting) is recommended as the most cost-effective solution that addresses the root cause. The grouting program should be designed with pre-grouting and post-grouting Lugeon testing to quantify improvement. Success criteria: reduce seepage to below 20 gpm and stabilize P-3 pore pressure below 12 psi.

3.0 ARCH CONCRETE — ASR MITIGATION

3.1 Condition Summary

ASR has degraded the outer 4 to 6 feet of the downstream arch face, reducing compressive strength by 26 percent and tensile strength by 37 percent. The interior concrete (75 percent of the arch thickness) is unaffected. The ASR reaction front is advancing at approximately 0.3 to 0.5 feet per decade based on the petrographic analysis. At this rate, 50 percent of the arch thickness would be affected in approximately 80 to 130 years, but strength degradation is expected to plateau as alkali supply is consumed.

3.2 Alternatives Evaluated

Alternative A: Lithium-Based Surface Treatment (Recommended)

Apply a penetrating lithium nitrate or lithium carbonate solution to the downstream face. Lithium compounds chemically convert the expansive ASR gel to a non-expansive lithium-silicate product, slowing or halting further ASR progression. Application by low-pressure spray in multiple coats. Typically provides 3 to 5 feet of penetration.

- **Cost:** \$120,000 to \$180,000 (full downstream face, scaffold and rigging access)
- **Schedule:** 3 to 4 weeks (low-flow season, fall 2026)
- **Effectiveness:** Moderate. Slows ASR progression but does not restore lost strength. Requires re-application every 5 to 10 years.

Alternative B: Silane/Siloxane Moisture Barrier

Apply a breathable silane/siloxane water repellent to the downstream face to reduce moisture ingress, which is essential for ASR gel expansion. Does not chemically arrest ASR but limits water availability for swelling.

- **Cost:** \$80,000 to \$120,000
- **Effectiveness:** Low to moderate. Downstream face is exposed to precipitation, reducing coating longevity. Re-application every 3 to 5 years.

Alternative C: Slot Cutting (Stress Relief)

Cut vertical slots in the downstream face of the arch at contraction joints to relieve compressive stresses caused by ASR expansion. This is a proven technique for ASR-affected dams (e.g., Mactaquac Dam, Canada) but is expensive and technically complex.

- **Cost:** \$800,000 to \$1,200,000
- **Effectiveness:** High for stress relief, but requires detailed FEM analysis to determine slot dimensions and locations. Appropriate only if structural analysis shows the arch is approaching stress limits.

Recommendation: Alternative A (lithium surface treatment) combined with the detailed FEM thermal-stress analysis recommended in the PFR report. The FEM analysis will determine whether the current strength reduction is structurally significant and whether slot cutting (Alternative C) may be needed in the future. The lithium treatment is a low-risk, moderate-cost measure that will slow ASR progression while the structural analysis is completed.

4.0 RIGHT ABUTMENT — DISPLACEMENT MANAGEMENT

4.1 Condition Summary

The right abutment crest has displaced 3.07 inches downstream over 30 years at a constant rate of 0.20 inches per year. The 2025 investigation confirmed the displacement is caused by creep in the glacial till foundation, not a developing failure surface. Static factor of safety is 1.42 (above FERC minimum of 1.3). Seismic FS is 1.08 (marginally below FERC minimum of 1.1 for pseudo-static analysis).

4.2 Alternatives Evaluated

Alternative A: Enhanced Monitoring (Recommended — Near Term)

Continue monitoring with the newly installed inclinometer and existing survey monuments. Quarterly inclinometer readings will confirm the depth of creep displacement and detect any change in mechanism. Alert levels: (1) Normal: rate remains at 0.20 in/yr; (2) Alert: rate exceeds 0.30 in/yr for two consecutive readings; (3) Action: rate exceeds 0.40 in/yr or inclinometer shows shear zone development.

- **Cost:** \$12,000 per year (quarterly readings plus analysis)

Alternative B: Newmark Seismic Analysis

Perform a Newmark sliding block analysis to evaluate the seismic displacement of the right abutment under the Maximum Credible Earthquake (MCE) and Operating Basis Earthquake (OBE). This analysis uses the actual time history of ground motion rather than the simplified pseudo-static approach, and typically yields more favorable results for embankments with FS > 1.0.

- **Cost:** \$25,000 to \$40,000
- **Purpose:** Determine if the seismic case is actually acceptable (likely, given FS > 1.0 pseudo-static), which would defer the need for a buttress.

Alternative C: Downstream Buttress (Contingency)

Construct a rockfill buttress against the downstream slope of the right abutment embankment to increase the factor of safety. Estimated buttress dimensions: 30 feet high, 120 feet long, 4,500 cubic yards of compacted rockfill.

- **Cost:** \$350,000 to \$500,000
- **Trigger:** Displacement rate exceeds 0.40 in/yr, OR Newmark analysis shows unacceptable seismic displacement, OR inclinometer detects shear zone.

Recommendation: Alternative A (enhanced monitoring) immediately, combined with Alternative B (Newmark analysis) in 2027 to resolve the seismic FS question. Alternative C (buttress) should be held in reserve as a contingency, with design completed to the 30-percent level to enable rapid mobilization if triggered.

5.0 COST SUMMARY AND IMPLEMENTATION SCHEDULE

Table 5-1: Implementation Cost Summary

Item	Description	Est. Cost	Timing	Priority
1	Left abutment supplemental grouting	\$180K-\$250K	Summer 2026	HIGH
2	ASR lithium surface treatment	\$120K-\$180K	Fall 2026	HIGH
3	FEM thermal-stress analysis (arch)	\$60K-\$90K	Q1 2026	HIGH
4	Inclinometer monitoring (quarterly)	\$12K/yr	Ongoing	MEDIUM
5	Enhanced seepage monitoring (weekly)	\$8K/yr	Ongoing	MEDIUM
6	Piezometer monitoring (monthly)	\$4K/yr	Ongoing	MEDIUM
7	Newmark seismic analysis	\$25K-\$40K	2027	MEDIUM
8	Rt. abutment buttress (30% design)	\$30K-\$45K	2027	LOW
9	Buttress construction (if triggered)	\$350K-\$500K	TBD	CONTINGENCY

Total estimated cost (2026-2027): \$439,000 to \$645,000 for all recommended actions (Items 1-8), excluding the contingency buttress. Annual monitoring costs of approximately \$24,000 thereafter. If the buttress is triggered, the total program cost increases to \$789,000 to \$1,145,000.

Funding consideration: EBID should consider whether FERC Safety of Dams (SOD) funding may be available for Items 1 through 3 as dam safety modifications. The grouting and ASR treatment are remedial actions addressing conditions that could affect long-term dam safety and may qualify for federal cost-sharing under the SOD program.

Please contact the undersigned with any questions regarding this memorandum or the recommended remediation program. We are prepared to proceed with detailed design of the grouting program and lithium treatment specification upon authorization.

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